

Smooth Manifolds

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1 Basic notions of manifolds

Definition 1.1: Smooth real manifolds

Example $\mathbb{S}^n$ ¹ is a smooth manifolds.

First use the stereographic projection, choose the charts $U_1 := \mathbb{S}^1 - \{(0, \dots, 0, 1)\}$ and $U_2 := \mathbb{S}^n - \{(0, \dots, 0, -1)\}$.

2 Submanifolds

3 Vector fields on smooth manifolds

4 Fiber bundles

5 Differential forms on manifolds

From tangent spaces to differential of functions

5.1 Differential forms

5.1.1 Motivation and definitions

The ultimate goal is to study the integrations on smooth real manifolds. Our work here is simply a preparation. Intuitively, integration involves the following three parts: (1) partitioning a region into infinitely many tiny, rigid geometric bodies, (2) assigning a value to each body (such as computing its mass or flux), and (3) summing these values.

For (1), on a general smooth real manifold, what mathematical objects can serve as these "tiny rigid bodies"?

- In the 1-dim case, it is represented by a single tangent vector $v \in T_p M$ at a point p .
- In the k -dim case, it is represented by an oriented parallelepiped spanned by k tangent vectors $v_1, \dots, v_k \in T_p M$. These oriented geometric bodies spanned by tangent vectors naturally reside in the k -th exterior power of the tangent space, denoted as $v_1 \wedge \dots \wedge v_k \in \bigwedge^k T_p M$.

(2) Since our "geometric bodies" live in $\bigwedge^k T_p M$, the operation of "assigning a value" to them—a map that consumes a geometric body and outputs a real number—naturally belongs to its dual space. This is the k -th exterior power of the cotangent space, denoted as $T_p^* M$.

(3) If we only wished to measure at a single point p , the algebraic structure above would suffice. However, calculus requires us to integrate over regions or even the entire manifold. This means we cannot just prepare a measuring device at one point; we must place a ready-to-use evaluation map at every single point of the manifold M . To transition from this "pointwise" perspective to a "global" one, we bundle the evaluation spaces of all points together to form the exterior cotangent bundle $\bigwedge^k T^* M$.

From the practical perspective, there is one more step—taking a section (making a choice in $\bigwedge^k T^* M$ for each point $p \in M$). This means precisely that we *smoothly*² assign a specific evaluation map $\omega_p \in \bigwedge^k T_p^* M$ to every point p on the manifold. By constructing this globally distributed field of measurement, we finally establish the foundation required to perform integration on manifolds.

Definition 5.1: Differential forms on manifolds

Let M be a smooth real manifold. The k -th differential form of M , denoted $\Omega^k(M)$, is simply the smooth global section $\Gamma(M, \bigwedge^k T^* M)$. Namely,

$$\Omega^k(M) := \Gamma\left(M, \bigwedge^k T^* M\right)$$

¹ $\mathbb{S}^n := \{(x_1, \dots, x_{n+1}) : x_1^2 + \dots + x_{n+1}^2 = 1\}$

²This assignment or choice is not arbitrary. To make each ω_p works well together, it has to be smooth.

5.1.2 Funtoriality of differential forms

5.2 Operations

5.2.1 Interior multiplication

5.2.2 Exterior derivative

Definition 5.2: Exterior derivative over $\mathbb{R}^n$

The exterior derivative map $d : \Omega^k(\mathbb{R}^n) \rightarrow \Omega^{k+1}(\mathbb{R}^n)$ is defined as $\omega \mapsto d\omega$.
More precisely, for a k -form $\omega = \sum_I \omega_I dx^I$, we define:

$$d\omega := \sum_I d\omega_I \wedge dx^I$$

where I is a shorthand for the index $(i_1, i_2, \dots, i_k)$ with $i_1 < i_2 < \dots < i_k$, and $dx^I = dx^{i_1} \wedge \dots \wedge dx^{i_k}$.

Remark This definition can be seen as applying d to every components ω_I , $dx^{i_1}, \dots, dx^{i_k}$. Because of the property that $d^2 = 0$, this point of view is equivalent to our definition.

Corollary 5.1: Properties of exterior derivatives over $\mathbb{R}^n$

Linear over $\mathbb{R}$. Antiderivative Nilpotency

Proof: Let $\omega = \sum_J \omega_J dx^J$ and $\eta = \sum_J \eta_J dx^J$.

$$\begin{aligned} d(\omega + \eta) &= d\left(\sum_J (\omega_J + \eta_J) dx^J\right) \\ &= \sum_J d(\omega_J + \eta_J) \wedge dx^J \\ &= \sum_J (d\omega_J + d\eta_J) \wedge dx^J \\ &= \sum_J d\omega_J \wedge dx^J + \sum_J d\eta_J \wedge dx^J \\ &= d\omega + d\eta. \end{aligned}$$

It suffices to consider forms $\omega = \sum_I \omega_I dx^I$ and $\eta = \sum_J \eta_J dx^J$.

$$\begin{aligned} d(\omega \wedge \eta) &= d(\omega_I \eta_J dx^I \wedge dx^J) \\ &= d(\omega_I \eta_J) \wedge dx^I \wedge dx^J \\ &= (\eta_J d\omega_I + \omega_I d\eta_J) \wedge dx^I \wedge dx^J \\ &= (d\omega_I \wedge dx^I) \wedge (\eta_J dx^J) + \omega_I d\eta_J \wedge dx^I \wedge dx^J \\ &= d\omega \wedge \eta + (-1)^k (\omega_I dx^I) \wedge (d\eta_J \wedge dx^J) \\ &= d\omega \wedge \eta + (-1)^k \omega \wedge d\eta. \end{aligned}$$

Notice that passing $d\eta_J$ across the k -form dx^I introduces the sign factor $(-1)^k$.

Step 1: For a 0-form (smooth function) $u \in \Omega^0(\mathbb{R}^n)$,

$$\begin{aligned} d(du) &= d\left(\sum_i \frac{\partial u}{\partial x^i} dx^i\right) \\ &= \sum_{i,j} \frac{\partial^2 u}{\partial x^j \partial x^i} dx^j \wedge dx^i \\ &= \sum_{i < j} \left(\frac{\partial^2 u}{\partial x^j \partial x^i} - \frac{\partial^2 u}{\partial x^i \partial x^j} \right) dx^j \wedge dx^i = 0, \end{aligned}$$

by Clairaut's theorem (symmetry of second derivatives).

Step 2: For an arbitrary k -form $\omega = \sum_J \omega_J dx^J$,

$$\begin{aligned} d(d\omega) &= d\left(\sum_J d\omega_J \wedge dx^J\right) \\ &= \sum_J d(d\omega_J) \wedge dx^J + \sum_J (-1)^1 d\omega_J \wedge d(dx^J) \\ &= 0 + 0 = 0. \end{aligned}$$

□

We wish to define the exterior derivative on a smooth manifold M . We achieve this by defining it locally on charts and showing it is well-defined on overlaps.

Theorem 5.1: Existence and uniqueness of exterior derivative on a manifold

For any smooth manifold M , there exists a unique operator $d : \Omega^k(M) \rightarrow \Omega^{k+1}(M)$ satisfying the properties: linearity, Leibniz rule, $d^2 = 0$, and $df(X) = Xf$ for $f \in C^\infty(M)$.

Local Construction and Gluing: Let (U, φ) be a smooth chart on M , where $\varphi : U \rightarrow \tilde{U} \subset \mathbb{R}^n$. We define the local exterior derivative $d_U : \Omega^k(U) \rightarrow \Omega^{k+1}(U)$ via the pullback of the standard derivative $\tilde{d}$ on $\mathbb{R}^n$:

$$d_U \omega := \varphi^* \left(\tilde{d}((\varphi^{-1})^* \omega) \right)$$

To ensure this operator globally patches together, consider another overlapping chart (V, ψ) . On the intersection $U \cap V$, we must verify that $d_U \omega = d_V \omega$. By examining the transition map $\psi \circ \varphi^{-1}$ and using the fact that the exterior derivative on $\mathbb{R}^n$ commutes with pullbacks naturally, we obtain:

$$\varphi^* \circ \tilde{d} \circ (\varphi^{-1})^* \omega = \psi^* \circ \tilde{d} \circ (\psi^{-1})^* \omega \quad \text{on } U \cap V$$

This compatibility allows us to uniquely glue the local operators d_U into a global operator d on M .

5.2.3 Functoriality

Corollary 5.2: Functoriality of exterior derivative

Let $F : M \rightarrow N$ be a smooth map. For a given k , $F^* : \Omega^k(N) \rightarrow \Omega^k(M)$ commutes with the exterior derivatives $d : \Omega^k(M) \rightarrow \Omega^{k+1}(M)$ and $d' : \Omega^k(N) \rightarrow \Omega^{k+1}(N)$. (the following diagram commutes)

$$\begin{array}{ccc} \Omega^k(N) & \xrightarrow{F^*} & \Omega^k(M) \\ d' \downarrow & & \downarrow d \\ \Omega^{k+1}(N) & \xrightarrow{F^*} & \Omega^{k+1}(M) \end{array}$$

Namely, $F^* \circ d' = d \circ F^*$. Elementwise, for any $\omega \in \Omega^k(N)$,

$$F^*(d' \omega) = d(F^* \omega)$$

Proof:

□

For any given k , $\Omega^k : \mathbf{Diff}_{\mathbb{R}} \rightarrow \mathbf{fdVect}_{\mathbb{R}}$ is a contravariant functor from the category of smooth real manifolds to real vector spaces. Since d assigns a morphism between the functors Ω^k and Ω^{k+1} making the diagram in 5.2 commutes, the exterior derivative d is a natural transformation.

5.2.4 Differential forms as sheaves

Lemma 5.0.1 (Cartan) Let M be a smooth manifold. If $\omega^1, \dots, \omega^k \in \Omega^1(M)$ are linearly independent 1-forms, and $\alpha^1, \dots, \alpha^k \in \Omega^1(M)$ are 1-forms satisfying:

$$\sum_{i=1}^k \alpha^i \wedge \omega^i = 0$$

Then there exist smooth functions $f_{ij} \in C^\infty(M)$ such that $f_{ij} = f_{ji}$ and

$$\alpha^i = \sum_{j=1}^k f_{ij} \omega^j$$

Remark (Intuition)

If the ω^i are linearly independent, they span non-degenerate oriented volumes. The cancellation $\sum \alpha^i \wedge \omega^i = 0$ implies that their magnitude and directions exactly cancel out. This forces every α^i to be linearly spanned purely by the basis formed by $\{\omega^1, \dots, \omega^k\}$.

Corollary 5.3: Cartan's magic formula

$$\mathcal{L}_X = i_v \circ d + d \circ i_v$$

Remark This formula is in fact saying that $\mathcal{L}_X$ is homotopic to the 0 maps, illustrated by the following diagram:

$$\begin{array}{ccccccc} \dots & \xrightarrow{d} & \Omega^k(M) & \xrightarrow{d} & \Omega^{k+1}(M) & \xrightarrow{d} & \dots \\ & \swarrow i_v & \downarrow \mathcal{L}_X & \swarrow i_v & \downarrow \mathcal{L}_X & & \\ \dots & \xrightarrow{d} & \Omega^k(M) & \xrightarrow{d} & \Omega^{k+1}(M) & \xrightarrow{d} & \dots \end{array}$$

But, the intuition is **unclear** for me. **What is the intuition for d and i_v and the coefficients ω_I for ω ?**

5.2.5 Application to $\mathbb{R}^3$ vector calculus

The machinery of differential forms elegantly generalizes classical vector calculus. Let $\mathfrak{X}(\mathbb{R}^3)$ denote the space of vector fields. We utilize the index-lowering (flat) isomorphism $\flat : \mathfrak{X}(\mathbb{R}^3) \xrightarrow{\sim} \Omega^1(\mathbb{R}^3)$ and its inverse (sharp) $\sharp$, alongside the Hodge star operator $*$.

For a vector field $X = P \frac{\partial}{\partial x} + Q \frac{\partial}{\partial y} + R \frac{\partial}{\partial z}$, we have:

$$\omega^1 = X^\flat = Pdx + Qdy + Rdz \in \Omega^1(\mathbb{R}^3)$$

$$\omega^2 = *(X^\flat) = Pdy \wedge dz + Qdz \wedge dx + Rdx \wedge dy \in \Omega^2(\mathbb{R}^3)$$

(Note: We use the cyclic permutation $dy \wedge dz, dz \wedge dx, dx \wedge dy$ to match the standard orientation).

Applying the exterior derivative:

$$\begin{aligned} d(X^\flat) &= \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) dy \wedge dz + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) dz \wedge dx + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx \wedge dy \\ &= (*\text{curl}(X))^\flat \end{aligned}$$

$$\begin{aligned} d(*X^\flat) &= \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \right) dx \wedge dy \wedge dz \\ &= (\text{div} X) dx \wedge dy \wedge dz = *(\text{div} X) \end{aligned}$$

This reveals the profound connection: the de Rham complex on $\mathbb{R}^3$ is completely isomorphic to the classical operators gradient, curl, and divergence. The property $d^2 = 0$ corresponds exactly to $\text{curl}(\text{grad} f) = 0$ and $\text{div}(\text{curl} X) = 0$. This diagram is commutative and $\text{curl} \circ \text{grad} = 0$, $\text{div} \circ \text{curl} = 0$.

$$\begin{array}{ccccccc} C^\infty(\mathbb{R}^3) & \xrightarrow{\text{grad}} & \mathfrak{X}(\mathbb{R}^3) & \xrightarrow{\text{curl}} & \mathfrak{X}(\mathbb{R}^3) & \xrightarrow{\text{div}} & C^\infty(\mathbb{R}^3) \\ \downarrow \text{id} & & \downarrow \flat & & \downarrow \sharp & & \downarrow * \\ \Omega^0(\mathbb{R}^3) & \xrightarrow{d} & \Omega^1(\mathbb{R}^3) & \xrightarrow{d} & \Omega^2(\mathbb{R}^3) & \xrightarrow{d} & \Omega^3(\mathbb{R}^3) \end{array}$$

Figure 1: Relations between different operators in vector calculus

6 Orientations on manifolds

6.1 Orientations on vector spaces and modules

6.1.1 Orientations on $\mathbb{R}$ -vector spaces

6.1.2 Orientations on free R -modules- a generalization

From now on, R is a commutative ring.

In the classical setting, an orientation on an n -dimensional $\mathbb{R}$ -vector space V (assume $\dim_{\mathbb{R}} V = n$) is defined via an ordered basis. Every such space admits exactly two orientations, which is equivalent to saying that the top exterior power $\bigwedge^n V$ is isomorphic to $\mathbb{R}$.

Now, we try to generalize orientations in two perspectives: (1) from a basis-dependent definition to a basis-free one. (2) from $\mathbb{R}$ -Vector Spaces to free R -Modules.

(1) Why seek a basis-free definition? Primarily to capture the intrinsic geometric nature of "handedness" without relying on arbitrary coordinates. Because we want to know if "orientation" is a fundamental algebraic property or just a shadow cast by the topology of real numbers. [more](#)

In exterior algebra, we represent oriented volumes (parallelepipeds) as elements of $\bigwedge^n V$. Since $\dim_{\mathbb{R}} V = n$, $\bigwedge^n V$ is a 1-dimensional $\mathbb{R}$ -vector space $\cong \mathbb{R}$. If we remove the zero element, the space $\bigwedge^n V \setminus \{0\}$ naturally splits into exactly two connected components. An orientation of V is simply a choice of one of these two connected components which are homeomorphic to two parts of the multiplicative groups $\mathbb{R}^\times$, namely $\mathbb{R}_{>0}$ and $\mathbb{R}_{<0}$.

(2) Can we extend this to a free R -module A over an arbitrary ring R ?

If we attempt to generalize via topology (as connected components), we encounter two major hurdles:

- Many rings lack a natural topology, making "connectedness" undefined.
- Even for topological rings like the p -adic integers $\mathbb{Z}_p$, the definition fails. Since $\mathbb{Z}_p$ is totally disconnected, the unit group $\mathbb{Z}_p^\times$ contains infinitely many "parts." This would lead to an infinite number of orientations, rendering the concept meaningless or pathological.

Therefore, defining orientation through the lens of topological connected components is not canonical; rather, it is an accidental property specific to $\mathbb{R}$. Instead, we turn to generalizing the orientation of an $\mathbb{R}$ -vector space using purely algebraic language. Since every ring possesses an algebraic structure, this approach is far more bound to work universally.

The central question becomes: what algebraic object associated with $\mathbb{R}$ naturally consists of exactly 'two parts'?

In $\mathbb{R}^\times$, the squares of all non-zero real numbers are strictly positive, implying that $(\mathbb{R}^\times)^2 = \mathbb{R}_{>0}$. Therefore, one natural candidate is the quotient group $\mathbb{R}^\times / (\mathbb{R}^\times)^2$. This provides an alternative perspective for dividing $\mathbb{R}^\times$ into a 'positive part' and a 'negative part'. Since $(\mathbb{R}^\times)^2 = \{x^2 \mid x \in \mathbb{R}^\times\}$, we have:

$$\mathbb{R}^\times / (\mathbb{R}^\times)^2 = \{1 \cdot (\mathbb{R}^\times)^2, -1 \cdot (\mathbb{R}^\times)^2\} \cong \{\pm 1\}$$

This framework transitions seamlessly to $\mathbb{Z}$ and the p -adic integers $\mathbb{Z}_p$. For instance, notice that $\mathbb{Z}^\times / (\mathbb{Z}^\times)^2 \cong \{\pm 1\}$. Consequently, every free $\mathbb{Z}$ -module (or free abelian group) retains exactly two possible orientations under this framework. The case for $\mathbb{Z}_p$ is much richer:

For odd primes p , we have the isomorphism $\mathbb{Z}_p^\times / (\mathbb{Z}_p^\times)^2 \cong \mathbb{F}_p^\times / (\mathbb{F}_p^\times)^2$. Since $\mathbb{F}_p^\times$ is a cyclic group of even order $p - 1$, the quotient modulo its subgroup of squares is a group of order 2. This quotient, $\mathbb{F}_p^\times / (\mathbb{F}_p^\times)^2$, is well-known as the dichotomy of quadratic residues and quadratic non-residues. [more](#) It is fascinating that free $\mathbb{Z}_p$ -modules are so special that their orientations can be represented by these classical number-theoretic classes. We will discuss this in more detail later.

When $p = 2$, the situation changes:

$$\mathbb{Z}_2^\times / (\mathbb{Z}_2^\times)^2 \cong \mathbb{Z}_2^\times / (1 + 8\mathbb{Z}_2) \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$$

As is often the case in number theory, $p = 2$ acts as a pathological case. This isomorphism reveals that any free $\mathbb{Z}_2$ -module should actually possess exactly four possible orientations.

Thus, the counterexample of free $\mathbb{Z}_2$ -modules shatters the naive intuition that 'every free R -module must have two orientations'. We are now ready to formally define an orientation on a free R -module A over an arbitrary commutative ring R :

An **orientation** of a free R -module A is an element of the quotient group $R^\times / (R^\times)^2$.

There is a pitfall. This definition is not intrinsic(canonical). Because we implicitly relies on an isomorphism

$$\bigwedge^m A \xrightarrow{\sim} R^\times$$

which is not canonical. We need to choose a basis first.

Suppose that $\text{rank}_R(A) = m$. To get a canonical definition, is it possible to directly choose $(\bigwedge^m A)^\times / ((\bigwedge^m A)^\times)^2$ as an intrinsic substitution of $R^\times / (R^\times)^2$? This makes no sense for the following reasons:

- $\bigwedge^m A$ is an R -module, not a ring. There is no unit elements $(\bigwedge^m A)^\times$. **really?**
- It makes no sense to use the square $((\bigwedge^m A)^\times)^2$, because every element $\omega \wedge \omega = 0$ degenerates for $\omega \in (\bigwedge^m A)^\times$.

Notice that: First, to define an orientation, only the multiplication of R is needed – $R^\times$ as a multiplication group. The obstruction of $\bigwedge^m A$ is that it has too many structures. $\bigwedge^m A$ is of rank 1. How about discarding the addition part of $\bigwedge^m A$? So, let's take the *torsor* of $\bigwedge^m A$, denoted $\text{Gen}(\bigwedge^m A)$, which is the set of all generators of $\bigwedge^m A$. $\text{Gen}(\bigwedge^m A)$ is in fact a substitution of $R^\times$. The idea behind it is applying a forgetful functor Gen ,

$$\text{Gen} : \mathbf{RMod} \rightarrow \mathbf{Gp}$$

There is a connection between $\bigwedge^m A$ and $R^\times$ via a group action

$$R^\times \curvearrowright \text{Gen}(\bigwedge^m A) \quad r \cdot \omega := r \cdot \omega$$

This group action also induces a group action of $(R^\times)^2$, $(R^\times)^2 \curvearrowright \text{Gen}(\bigwedge^m A)$. The orbit of this action is defined to be the collection of all orientation and each orbit is a orientation, which is the canonical substitution of $R^\times / (R^\times)^2$. So,

Definition 6.1: Orientation for a free R -module

Let R be a commutative ring and A be a free R -module of finite rank.
Each element in the orbit $\text{Orb}_{(R^\times)^2}(\text{Gen}(\bigwedge^m A))$ is an orientation of A .

Remark In this definition, orientations merely use the multiplication of R . Is it possible to extend the notion of orientations simply to group-related objects, such as G -modules? This is not possible because the exterior structure requires A to be an R -module.

Generalizing orientations to arbitrary free R -modules over any ring R is therefore not merely "abstract nonsense," but a deeply meaningful construction. **what Equivalency of definitions**

Morphisms between two free R -modules pertaining to orientations

Having established the notion of orientation for a single free R -module A , the natural next step is to examine morphisms between two free R -modules, A and B .

Let $f : A \rightarrow B$ be an R -linear map. By the functoriality of the exterior algebra, f induces a morphism between exterior products $\bigwedge f : \bigwedge A \rightarrow \bigwedge B$. Let us assume that $\text{rank}_R A = m$ and $\text{rank}_R B = n$.

(1) First, consider the case that $m = n$ so that we may meaningfully compare their top exterior powers. Given that both A and B are equipped with an orientation, how does f interact with these structures? Does it map the orientation of A into the orientation of B , or does it fail to preserve it? (In the classical case where $|R^\times / (R^\times)^2| = 2$, "not preserving" simply means "reversing").

To delve into this problem, it is easier to think about an endomorphism of A , $f : A \rightarrow A$. Still, A is a free R -module of rank m . Let $E = \{v_1, \dots, v_m\}$ be a basis of A , and let $E' = \{v'_1, \dots, v'_m\}$ be the set of their images, such that $v'_i = f(v_i)$. We define the top exterior generators as $\omega_E := v_1 \wedge \dots \wedge v_m$ and $\omega_{E'} := v'_1 \wedge \dots \wedge v'_m$.

The Invertible Case: If f is an isomorphism, E' remains a basis of A , and $\det(f) \in R^\times$. Since the induced map on the top exterior power yields $(\bigwedge^m f)(\omega_E) = \omega_{E'}$ and $(\bigwedge^m f)(\omega_E) = \det(f) \cdot \omega_E$, we have $\omega_{E'} = \det(f) \cdot \omega_E$. Thus, $\omega_{E'}$ is merely a unit multiple of ω_E , allowing us to discuss orientation preservation or reversal.

The Degenerate Case: If f is not invertible, E' fails to be a basis. The relation $\omega_{E'} = \det(f) \cdot \omega_E$ still holds, but now $\det(f) \in R \setminus R^\times$. Consequently, $\det(f)$ does not generate the unit ideal in R , meaning that the image $R \cdot \omega_{E'}$ is a proper submodule of $\bigwedge^m A$. The orientation is essentially crushed. To understand the algebraic "defect" caused by this collapse, consider the cokernel $A/f(A)$. For any commutative ring R , utilizing the adjugate map (classical adjoint) $f^* : A \rightarrow A$, we have the identity $f \circ f^* = f^* \circ f = \det(f) \cdot \text{id}_A$. Therefore, for any $x \in A$:

$$\det(f) \cdot x = f(f^*(x)) \in f(A)$$

This proves that the quotient $A/f(A)$ is annihilated by $\det(f)$, or $\det(f) \in \text{Ann}(A/f(A))$.

We can view this elegantly through a functorial lens, comparing the objects and morphisms in the category of modules with those in their top exterior powers:

$$\begin{array}{ccc} A & \xrightarrow{\wedge^m} & \wedge^m A \\ \downarrow f & & \downarrow \wedge^m f \\ f(A) & \xrightarrow{\wedge^m} & \text{im}(\wedge^m f) \end{array}$$

Note that the image of the induced map is $\text{im}(\wedge^m f) = \wedge^m f(R \cdot \omega_E) = R \cdot (\det(f) \cdot \omega_E)$. This allows us to construct the isomorphism of the quotients:

$$\wedge^m A / \text{im} \left(\wedge^m f \right) \cong \frac{R \cdot \omega_E}{R \cdot (\det(f) \cdot \omega_E)} \cong R / \langle \det(f) \rangle$$

This quotient ring precisely measures the "defect" of f , or how far f is from being invertible. If f is invertible, $\det(f) \in R^\times$, generating the whole ring ($\langle \det(f) \rangle = R$), making the quotient $R/R \cong \{0\}$. If f is not invertible, $\det(f)$ lies outside $R^\times$, making $\langle \det(f) \rangle$ a proper ideal of R . The structure of $R/\langle \det(f) \rangle$ encodes exactly how the mapping crushes the volume and destroys the orientation of the module.

To formalize this, we can denote the oriented modules as pairs $(A, [\omega_A])$ and $(B, [\omega_B])$, where $\omega_A \in \wedge^n A$ and $\omega_B \in \wedge^n B$ are basis generators. Their equivalence classes, $[\omega_A]$ and $[\omega_B]$ modulo $(R^\times)^2$, represent the chosen orientations for A and B , respectively. Since ω_B generates the rank-1 free module $\wedge^n B$, there exists a unique scalar $c \in R$ such that: $c \in R^\times$?

$$\left(\wedge^n f \right) (\omega_A) = c \cdot \omega_B$$

If $c \in (R^\times)^2$, the map f is said to be orientation-preserving.

If $c \notin (R^\times)^2$, the map f alters the orientation (or does not preserve it).

Remark: Even though $\wedge^n A \cong R$, it is conceptually misleading to simply state that $\omega_A \in R^\times$. While we can interpret ω_A as an element of $R^\times$ because it acts as a generator for $\wedge^n A$, doing so is mathematically equivalent to canonically fixing an element in $R^\times$. Fixing this element dictates a specific isomorphism $\wedge^n A \xrightarrow{\sim} R$, which is precisely what it means to choose an orientation for A .

When $R = \mathbb{Z}_p$

So far we just discussed that $f : A \rightarrow B$ is an R -linear map without saying whether it is invertible or not.

In our previous discussion on morphisms between oriented free R -modules, we analyzed the coefficient $c \in R$. If $c \in R^\times$, the map modifies the orientation cleanly. However, what occurs when $c \notin R^\times$? This breaks the symmetry and leads us to the degenerate cases where the morphism is no longer an isomorphism, effectively "collapsing" the geometric structure.

What if $\text{rank}_R A = m$, $\text{rank}_R B = n$, and $m \neq n$? In this case, we lose the direct comparison of a single "top-dimensional volume".

Regardless of whether m or n is larger, the exterior algebra functor is globally defined. To preserve as much geometric information as possible, instead of taking the m -th or n -th exterior power, we should take their minimum:

$$k = \min(m, n)$$

We examine the induced map $\wedge^k f : \wedge^k A \rightarrow \wedge^k B$. This unified definition is the cornerstone for the subsequent case-by-case analysis.

(2) $m < n$ (Embedding and Extrinsic Posture):

Here, $k = m$. The map is $\wedge^m f : \wedge^m A \rightarrow \wedge^m B$. Geometrically, imagine placing a lower-dimensional space with an intrinsic orientation (like a 2D sheet of paper) into a higher-dimensional space. Its intrinsic orientation remains unchanged, but it acquires a "posture" or extrinsic orientation relative to the ambient space. At this point, $\wedge^m A$ is 1-dimensional, possessing a generator ω_A (representing the global orientation of A). However, $\wedge^m B$ is a higher-dimensional space. The result of the mapping, $(\wedge^m f)(\omega_A) = \sum c_J \cdot \omega_{B,J}$, is no longer a single scalar, but a linear combination. This set of coefficients $\{c_J\}$ (which are the $m \times m$ maximal minors of the transformation matrix) precisely quantifies the "projection shares" of A 's orientation onto the various coordinate planes of the higher-dimensional space.

(3) $m > n$ (Projection, Information Loss, and Sub-orientations)

Here, $k = n$. The map is $\bigwedge^n f : \bigwedge^n A \rightarrow \bigwedge^n B$. Projecting a higher-dimensional object onto a lower-dimensional space. The global higher-dimensional orientation (e.g., a 3D volume) suffers a "dimensional collapse" and is completely destroyed (since $\bigwedge^m B = 0$). Although the global orientation is dead, A contains infinitely many n -dimensional subspaces (slices). We select a subset of basis elements from A indexed by $I = \{i_1, \dots, i_n\}$, which span a submodule A_I of rank n .

Local Return to the $m = n$ Case: We restrict the morphism to these subspaces, yielding $f_I : A_I \rightarrow B$. Now, the ranks on both sides are equal! Conclusion: When the orientation ω_I of each subspace A_I is mapped over, it produces a scalar multiple $c_I \cdot \omega_B$. This c_I is the determinant of the restricted map, $\det(f_I)$! It turns out that a morphism where $m > n$ essentially slices the higher-dimensional space and computes the determinant ($n \times n$ maximal minor) for each slice.

R S

6.1.3 Orientations on manifolds

Now, we shift our perspective from orientations on abstract vector spaces to orientations on manifolds. Our goal is to rigorously formulate a notion of orientation for the latter. To study the orientation of a manifold, we must identify vector spaces naturally associated with it. A naive guess for a smooth n -dimensional manifold M might be to utilize $\mathbb{R}^n$ itself. By definition, M is locally homeomorphic to $\mathbb{R}^n$, meaning that every point $p \in M$ admits a neighborhood U equipped with a local homeomorphism (a coordinate chart) $\varphi : U \rightarrow \mathbb{R}^n$. However, relying on these local homeomorphisms to $\mathbb{R}^n$ introduces several fundamental drawbacks:

- Lack of Algebraic Transfer: The map φ is merely a homeomorphism; it does not preserve the algebraic vector space structure of $\mathbb{R}^n$. More precisely, the pullback φ^{-1} cannot endow the open set U with a vector space structure. $\mathbb{R}^n$ acts merely as an extrinsic topological attachment, a concomitant coordinate space rather than an intrinsic object of M .

- Coordinate Dependence: The chart φ automatically imposes a specific basis $\varphi = (x^1, x^2, \dots, x^n)$. While this induces a basis for the tangent space $T_p M$ at any point p (namely, the derivation operators $\{\frac{\partial}{\partial x^1}|_p, \dots, \frac{\partial}{\partial x^n}|_p\}$), the very presence of φ directly contradicts our primary objective of establishing a strictly coordinate-free theory.

- Field Bias (The Arithmetic Obstruction): Anchoring the definition to charts would inextricably bind the orientation of a manifold to the field of real numbers $\mathbb{R}$. This association heavily biases our intuition toward the specific analytical properties of $\mathbb{R}$, thereby hindering the development of a truly general theory for manifolds over arbitrary fields (such as p -adic manifolds or algebraic varieties). We must strictly decouple our geometric intuition from the arithmetic of $\mathbb{R}$ to achieve a canonical, universal understanding of manifolds. The only truly intrinsic objects of a manifold that simultaneously carry a vector space structure are its tangent spaces. This realization naturally leads us to a localized definition.

Definition 6.2: Pointwise Orientation of a Manifold

A pointwise orientation of a smooth manifold M is defined as a collection of choices of orientations for every tangent space of M :

$$\mathcal{O} = \{\omega_{T_p M} \mid p \in M\}$$

where each $\omega_{T_p M}$ represents an orientation class in $\bigwedge^n T_p M$.

Remark This formulation, however, remains inadequate. The orientation of a manifold ought to be a global, cohesive topological property. Under this strictly pointwise definition, the chosen orientations for two infinitesimally nearby tangent spaces could be wildly discontinuous and irrelevant to one another.

To build geometric intuition for what constitutes a proper global orientation on a manifold, consider the classic pathological example: the Möbius strip. Starting at a point $p \in M$, we can locally assign an orientation $\omega_{T_p M}$ to the tangent space $T_p M$. We can then continuously extend this choice to the tangent spaces of nearby points, enforcing "local consistency." A rigorous global orientation demands that this consistency holds everywhere, preventing any single tangent space from accommodating conflicting orientation assignments. The Möbius strip fundamentally violates this requirement. Regardless of the initial pointwise orientation chosen, maintaining "nearby consistency" while traversing the central loop results in a topological obstruction: upon returning to the starting point p , the continuously transported orientation diverges from the original $\omega_{T_p M}$, becoming its exact opposite. This geometric obstruction motivates rigorous approaches to formalize global orientability. The text highlights the following pathways:

1. The Atlas Approach (Transition Functions) A smooth manifold M is orientable if it admits an atlas

$\{(U_\alpha, \varphi_\alpha)\}$ such that for any two overlapping charts $(U_\alpha \cap U_\beta \neq \emptyset)$, the Jacobian determinant of the transition function is strictly positive:

$$\det(d(\varphi_\beta \circ \varphi_\alpha^{-1})) > 0$$

This ensures that the local algebraic structure (the "handedness" of $\mathbb{R}^n$) is coherently patched together across the entire manifold.

The Monodromy/Holonomy Approach (The Fundamental Group) Our intuition suggests that global inconsistency is inherently a macroscopic phenomenon detected by traversing closed paths. Locally, inconsistencies can always be hidden or rectified by adjusting the local charts. This leads to the viewpoint of monodromy or holonomy along loops. Given a closed curve $\gamma : [0, 1] \rightarrow M$ with $\gamma(0) = \gamma(1) = p$, we can continuously transport an orientation of the tangent spaces along the image of γ . As γ is fully traversed from $t = 0$ to $t = 1$, the resulting transported orientation at $T_{\gamma(1)}M$ either strictly agrees with the initial orientation at $T_{\gamma(0)}M$ (assigned a value of $+1$), or it is reversed (assigned a value of -1). A Crucial Subtlety: The existence of some orientation-preserving curve is completely inadequate to prove global orientability. For instance, on a Möbius strip, traversing the central loop twice produces a curve that successfully preserves the orientation ($\omega_{T_{\gamma(1)}M} = \omega_{T_{\gamma(0)}M}$). Therefore, orientability must be a universal property, holding consistently for arbitrary closed loops. To systematically encode the behavior of all possible closed loops, we must examine the fundamental group $\pi_1(M, p)$. This holonomy behavior naturally induces a group homomorphism from the fundamental group to $\mathbb{Z}/2\mathbb{Z}$:

$$w_1 : \pi_1(M, p) \rightarrow \mathbb{Z}/2\mathbb{Z}$$

A manifold M is said to be orientable if and only if w_1 is the trivial homomorphism. The triviality of w_1 guarantees that the orientation is preserved along all closed loops. This homomorphism w_1 is a fundamental topological invariant, classically known as the first Stiefel-Whitney class (often viewed in cohomology as $w_1 \in H^1(M; \mathbb{Z}/2\mathbb{Z})$).

The Orientation Bundle Approach (Continuous Sections) The third viewpoint conceptualizes the pointwise assignment of orientations as a global mathematical function. One might consider a map from M into the collection of all possible orientations across all tangent spaces. However, simply assigning a value at each point is insufficient. On a non-orientable surface like the Möbius strip, forcing a global assignment inevitably causes the orientation value to abruptly "jump" or reverse after traversing the central loop. Therefore, to rigorously capture 'orientability', this assignment function must be strictly continuous. To formalize this, we define the disjoint union of all pointwise orientations:

$$\widetilde{M} = \coprod_{x \in M} \mathcal{O}_x$$

where $\mathcal{O}_x$ is the two-element set of orientations for $T_x M$. We endow $\widetilde{M}$ with a natural topology such that it forms a double cover (a 2-fold covering space, or a principal $\mathbb{Z}/2\mathbb{Z}$ -bundle) over M . This space $\widetilde{M}$ is called the orientation bundle. There is a natural covering projection $\pi : \widetilde{M} \rightarrow M$.

Definition 6.3: Global orientation on a manifold(bundle approach)

A global orientation on M is defined as a continuous global section of π . That is, there exists a continuous map $s : M \rightarrow \widetilde{M}$ such that $\pi \circ s = \text{id}_M$.

The three distinct perspectives discussed above (Transition Functions, Monodromy/Stiefel-Whitney, and Orientation Bundles) are entirely equivalent definitions for orientability on M . Regardless of the chosen perspective, M is orientable if at least one global orientation exists. Thus, an oriented manifold is formally defined as an ordered pair $(M, \mathcal{O})$, where M is an orientable manifold and $\mathcal{O}$ is a specific choice of global orientation. The orientation induced on $T_p M$ by $\mathcal{O}$ is typically denoted as $\mathcal{O}_p$. (Generalization Note): An oriented manifold with boundary is defined in a completely analogous manner.

Furthermore, for a 0-dimensional smooth manifold (which is simply a discrete set of points), an orientation is conventionally defined as a choice of a sign (± 1) attached to each point.

7 Integrations on manifolds

8 De Rham cohomology

8.1

Let M be a real smooth manifold. In classical expression of de Rham cohomology, we study the chain complex

$$0 \rightarrow \Omega^0(M) \rightarrow \Omega^1(M) \rightarrow \Omega^2(M) \rightarrow \dots \rightarrow \Omega^k(M) \rightarrow \Omega^{k+1}(M) \rightarrow \dots \quad (1)$$

Since we mentioned in section that

$$\Omega^\bullet : 0 \rightarrow \Omega^0 \rightarrow \Omega^1 \rightarrow \dots \rightarrow \Omega^k \rightarrow \Omega^{k+1} \rightarrow \dots \quad (2)$$

is a *long exact sequence* of sheaves (not only a chain complex) due to the Poincaré lemma. This long exact sequence is also a resolution of a constant sheaf $\underline{\mathbb{R}}$.

Therefore, we have a functor from the category of sheaf $\mathbf{Sh}(M, \mathbf{Ring})$ to the category of complex $\mathbf{Comp}(\mathbf{Sh}(M, \mathbf{Ring}))$. How should we get 2 from 1 by applying some functor? The standard way is to use the global section functor $\Gamma(-, M)$. Hence, for the de Rham cohomology group, there is a reformulation in the language of derived functors:

$$\boxed{H_{\text{dR}}^k(M) := H^k(\Omega^\bullet) = R^k(\Gamma(M, -))(\underline{\mathbb{R}})}$$

which is also known as the abstract de Rham theorem. The chain $\Omega^\bullet$ is exact, but after applying the global section functor $\Gamma(M, -)$, the resulting chain complex has no guarantee of being exact in general.

The whole process is illustrated in the following diagram

$$\begin{array}{c} \mathbf{Sh}(M, \mathbf{Ring}) \\ \downarrow \\ \mathbf{Comp}(\mathbf{Sh}(M, \mathbf{Ring})) \\ \downarrow \Gamma(M, -)^\bullet \\ \mathbf{Comp}(\mathbf{fdVect}_{\mathbb{R}}) \\ \downarrow \\ \mathbf{fdVect}_{\mathbb{R}} \end{array}$$

Figure 2: The Abstract de Rham Theorem Process

$\Gamma(M, -)^\bullet$ is the functor between two chain complex categories induced by $\Gamma(M, -)$.